

Trustworthy AI Assignment 1

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Abstract—This report analyzes adversarial attacks on MNIST and CIFAR-10 using targeted and untargeted FGSM and PGD. The report includes configuration ablation and epsilon sweep analysis, and further examines model behavior with t-SNE and saliency map visualization. Code is available at: <https://github.com/ANHANYOUNG/trustworthy-ai-assignment1>.

I. INTRODUCTION

This assignment evaluates targeted and untargeted FGSM and PGD attacks on MNIST and CIFAR-10. The report includes configuration ablation, epsilon sweep analysis, and additional visualization in Appendix A and Appendix B.

II. EXPERIMENTAL SETUP

Experiments were conducted on MNIST with a CNN classifier and on CIFAR-10 with a ResNet-based classifier [1]. I evaluated targeted and untargeted FGSM [2] and PGD [3], and report clean accuracy, attack success rate, and adversarial accuracy. The MNIST classifier was trained for 3 epochs and the CIFAR-10 classifier for 10 epochs, both using Adam, CELoss, a batch size of 64, and a learning rate of 10^{-3} . In targeted attacks, the target class was fixed to 0. For the main evaluation, FGSM and PGD used $\epsilon = 0.3$ on MNIST and $\epsilon = 8/255$.

III. EXPERIMENTAL RESULTS

A. Configuration Ablation

Tables I and II summarize the configuration ablation results. In FGSM, targeted attacks did not reach full success for all settings, but untargeted attacks were successful in most cases. PGD was successful in almost all cases across configurations. The highest targeted FGSM attack success rate was observed for the model with dropout, without batch normalization, and with GELU activation (74.5%), while the lowest targeted FGSM attack success rate was observed for the model with dropout, with batch normalization, and with GELU activation (2.1%).

On CIFAR-10, the model without dropout, with batch normalization, and with GELU activation showed the highest FGSM attack success rates. In contrast, the model without dropout, without batch normalization, and with ReLU activation showed the lowest average attack success rate overall. A detailed explanation is provided in Section IV.

B. Epsilon Sweep on the Best Model

Based on the ablation study, the best clean model for both MNIST and CIFAR-10 was selected for additional epsilon sweep analysis. Tables III and IV report the attack success rates across different ϵ values.

TABLE I: Configuration ablation results on MNIST Blue indicates strong robustness, and Red indicates weak robustness. Since PGD is successful across almost all configurations, it is not highlighted.

Configuration			Clean (%)		FGSM Target		FGSM Untarget		PGD Target		PGD Untarget	
Drop	BN	Act			SR ↓	Acc ↑	SR ↓	Acc ↑	SR ↓	Acc ↑	SR ↓	Acc ↑
N	N	ReLU	99.03		12.7	22.2	91.5	8.5	99.7	0.0	100.0	0.0
N	N	GELU	98.94		55.3	2.4	97.8	2.2	100.0	0.0	100.0	0.0
N	Y	ReLU	99.07		24.1	4.6	98.1	1.9	100.0	0.0	100.0	0.0
N	Y	GELU	99.14		34.1	2.0	99.5	0.5	100.0	0.0	100.0	0.0
Y	N	ReLU	98.77		13.2	20.9	87.8	12.2	99.1	0.4	100.0	0.0
Y	N	GELU	99.11		74.5	0.6	97.7	2.3	100.0	0.0	100.0	0.0
Y	Y	ReLU	98.90		12.5	8.2	94.9	5.1	100.0	0.0	100.0	0.0
Y	Y	GELU	99.05		2.1	8.7	95.6	4.4	100.0	0.0	100.0	0.0

TABLE II: Configuration results on CIFAR-10 Blue indicates strong robustness, and Red indicates weak robustness. Since PGD is successful across almost all configurations, it is not highlighted.

Configuration			Clean (%)		FGSM Target		FGSM Untarget		PGD Target		PGD Untarget	
Drop	BN	Act			SR ↓	Acc ↑	SR ↓	Acc ↑	SR ↓	Acc ↑	SR ↓	Acc ↑
N	N	ReLU	78.40		9.8	26.1	85.9	14.1	94.6	1.6	99.9	0.1
N	N	GELU	81.16		19.4	16.9	89.1	10.9	99.9	0.0	100.0	0.0
N	Y	ReLU	87.81		9.7	11.2	95.6	4.4	100.0	0.0	100.0	0.0
N	Y	GELU	87.82		31.4	8.4	96.9	3.1	100.0	0.0	100.0	0.0
Y	N	ReLU	75.38		16.4	24.9	87.0	13.0	97.0	0.9	99.8	0.2
Y	N	GELU	80.29		10.9	20.5	84.1	15.9	98.0	0.4	99.2	0.8
Y	Y	ReLU	86.80		18.8	8.6	96.5	3.5	100.0	0.0	100.0	0.0
Y	Y	GELU	86.20		16.3	9.6	96.9	3.1	100.0	0.0	100.0	0.0

TABLE III: MNIST epsilon sweep attack success rates.

ϵ	FGSM-T (%)	FGSM-U (%)	PGD-T (%)	PGD-U (%)
0.05	1.0	8.4	1.8	17.8
0.10	5.6	37.0	45.8	91.8
0.20	31.8	93.8	100.0	100.0
0.30	34.4	99.4	100.0	100.0

For MNIST, attack success rates generally increased as ϵ increased. At the same perturbation level, untargeted attacks showed higher success rates than targeted attacks, and PGD showed higher success rates than FGSM. In particular, PGD reached 100% success from $\epsilon = 0.2$ in both targeted and untargeted settings, while FGSM increased more gradually.

For CIFAR-10, PGD again showed higher success rates than FGSM, and untargeted attacks generally showed higher success rates than targeted attacks. In targeted FGSM, the success rate increased from 1/255 to 2/255, but decreased at larger ϵ values. In contrast, targeted PGD reached nearly complete success from 2/255.

TABLE IV: CIFAR-10 epsilon sweep attack success rates.

ϵ	FGSM-T (%)	FGSM-U (%)	PGD-T (%)	PGD-U (%)
1/255	27.2	79.2	45.6	93.6
2/255	43.2	93.6	99.0	100.0
4/255	31.0	97.0	100.0	100.0
8/255	9.2	96.6	100.0	100.0

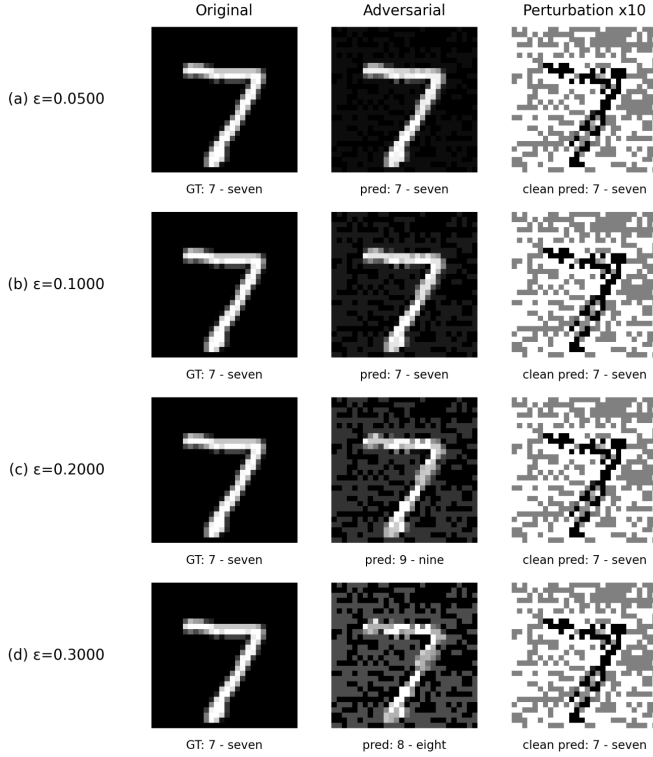


Fig. 1: Qualitative epsilon sweep for untargeted FGSM on MNIST. As ϵ increases, the perturbation becomes stronger and eventually changes the prediction, while the overall digit structure remains visually recognizable.

IV. DISCUSSION

PGD was generally stronger than FGSM because it refines the perturbation over multiple steps, while FGSM relies on a single update and is therefore more sensitive to gradient mismatch or overshooting. The ablation results show that robustness depends on the combination of dropout, batch normalization, and activation choice. Dropout did not provide a consistent robustness benefit, and batch normalization improved clean accuracy more reliably than adversarial robustness. GELU-based models were often more vulnerable to FGSM than ReLU-based models, possibly because the smoother nonlinearity of GELU produces gradients that are easier to exploit in one-step attacks.

The epsilon sweep revealed a trade-off between attack success and perturbation visibility: larger ϵ generally improved attack success but made perturbations easier to notice.

An unexpected result appeared in targeted FGSM on

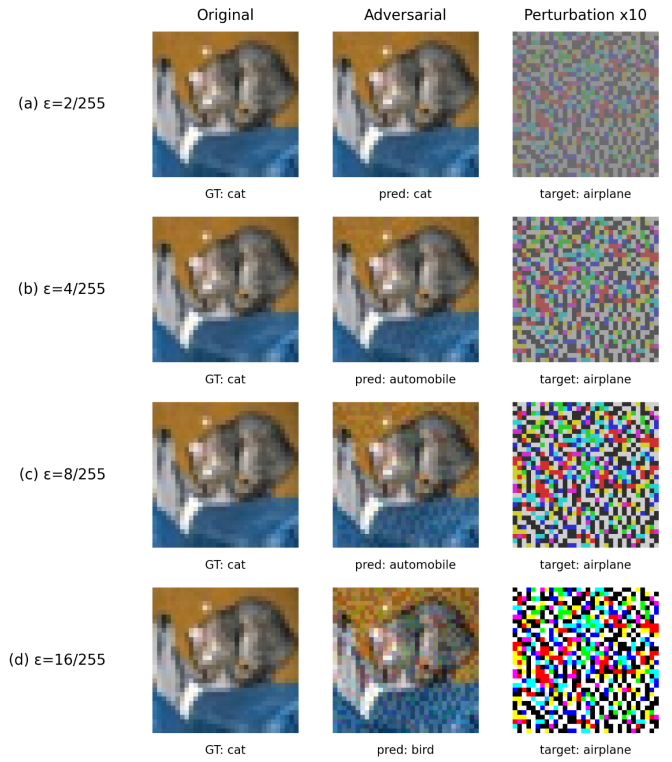


Fig. 2: Qualitative epsilon sweep for targeted FGSM on CIFAR-10. Larger ϵ values make the perturbation increasingly visible, but stronger one-step perturbations do not always lead to better targeted attack success.

CIFAR-10, where success did not increase monotonically with ϵ . A likely explanation is that a larger one-step targeted perturbation may overshoot the desired direction, and single-step targeted attacks are more sensitive to gradient mismatch because they cannot iteratively correct the update.

Additional qualitative analysis is provided in Appendix A and Appendix B.

V. CONCLUSION

This report evaluated targeted and untargeted FGSM and PGD attacks on MNIST and CIFAR-10. The results showed that attack success rates varied depending on attack type, perturbation size, and model configuration. In particular, PGD generally achieved higher success rates than FGSM, and larger ϵ values generally increased attack success while making perturbations more visible. The experiments also revealed an unexpected pattern in targeted FGSM on CIFAR-10, where increasing ϵ did not always improve attack success.

REFERENCES

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APPENDIX

A. t-SNE Visualization

t-SNE is used to visualize how adversarial perturbations change the feature distribution [4]. As shown in Fig. 3, targeted PGD produces stronger class mixing, while untargeted PGD yields a more scattered distribution. This suggests that targeted attacks push features toward a specific target-related region, whereas untargeted attacks mainly break the original class structure more broadly.

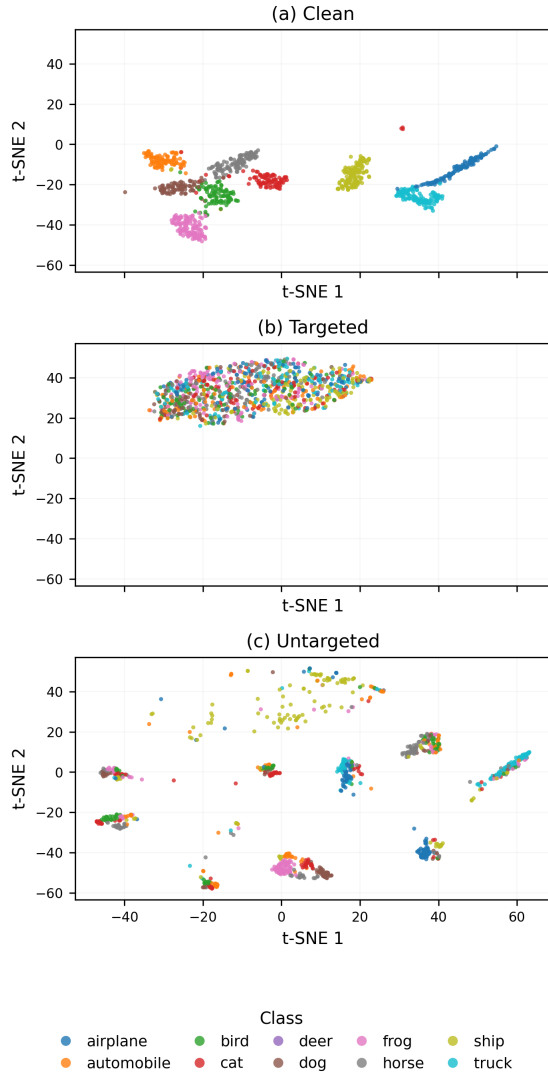


Fig. 3: t-SNE visualization of CIFAR-10 features under PGD.

B. Saliency Map Analysis

Saliency maps visualize how adversarial perturbations change the model's focus in the input space [5]. Fig. 4 shows different saliency patterns for clean and adversarial inputs. Compared with the clean inputs, the adversarial examples produce more dispersed and shifted saliency responses, indicating that the model attends to different pixel regions after perturbation. In the targeted setting, the saliency often appears more concentrated around features consistent with the target class, while in the untargeted setting the attention becomes broadly distorted without necessarily aligning to a single alternative class pattern.

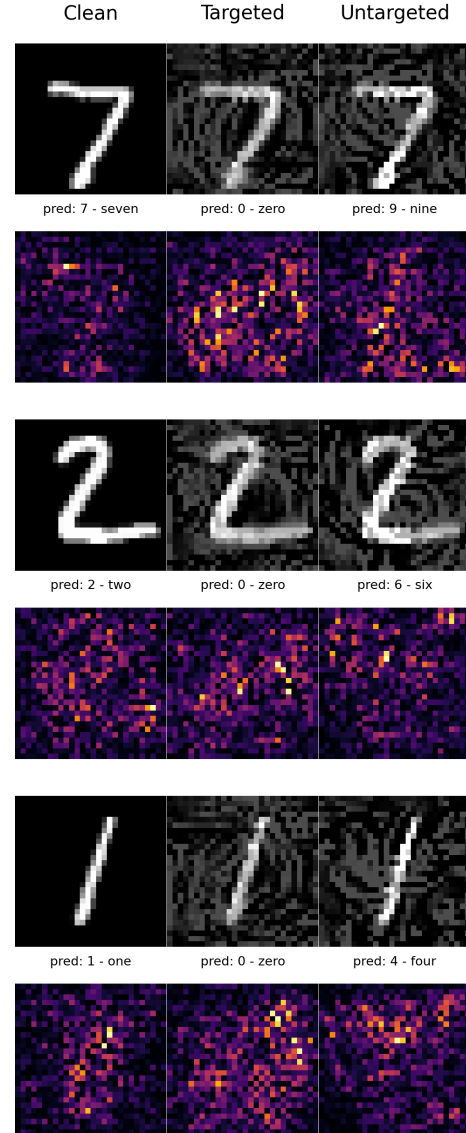


Fig. 4: Saliency map comparison for clean and adversarial inputs.